

llmXive Automated Review of MiniMax Sparse Attention

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: [agents/prompts/paper_reviewer_claim_accuracy.md](#)

The review focuses on the accuracy of factual claims and the alignment between claims and

cited evidence.

1. Overstated Performance Claims vs. Empirical Data In the Abstract and Introduction, the authors claim that MiniMax Sparse Attention (MSA) “performs on par with GQA” and “matches GQA on benchmarks.” While Table 1 (Main Results) shows MSA-PT and MSA-CPT performing comparably or slightly better on general reasoning (MMLU, MMLU-Pro) and retrieval (RULER-8K/32K), Table 2 (Long-context extension) presents a contradiction. In the HELMET-128K benchmark, the Full Attention model scores **46.53**, while the MSA-CPT model scores **45.93**, a drop of **-0.60**. The text describes this as “maintains performance,” but the data shows a measurable degradation. The claim of “matching” is too strong given this specific result. The text should be revised to state that performance is “comparable” or “largely preserved,” acknowledging the slight dip in long-context retrieval metrics.

2. Contradiction Between Mechanism Description and Ablation Results The Introduction states that the Index Branch “always retaining the most recent block” (local block) is a key design feature to ensure stability. However, the ablation study in Appendix A.2 (Table: Forced Sink & Local Selection) explicitly states: “Removing forced sink/local selection has little effect on quality; the model learns these patterns.” This suggests that the explicit mechanism of *forcing* the local block is not as critical as the text implies, as the model naturally learns to attend to local tokens. The claim that the mechanism is essential for stability is weakened by the ablation results. The text should be nuanced to reflect that while the mechanism is present, the model’s ability to learn local attention patterns makes the explicit forcing less of a hard requirement for quality than currently stated.

3. Citation Accuracy for Hybrid Architectures In Section 5 (Related Works), the text claims: “Hybrid stacks~\citep{minimax01,minimaxm1} interleave linear and full-attention blocks.” The citation \citep{minimax01} refers to “MiniMax-01: Scaling Foundation Models with Lightning Attention.” While “Lightning Attention” is a linear attention mechanism, the specific claim of “interleaving linear and full-attention blocks” is a structural detail that may not be the primary focus of the cited paper or may require a more specific citation to a paper that explicitly details this interleaving architecture. If `minimax01` does not explicitly describe an interleaved hybrid stack, this citation is inaccurate for the specific claim made. The authors should verify if `minimax01` supports the “interleaving” claim or if a different reference is more appropriate.

4. Kernel Speedup Claims The claim of “5.1× speedup vs \texttt{torch} at $N = 128K$ ” (Section 4.1) is supported by Table 1 (Top- k latency), which shows 3970 μs for \texttt{torch} vs. 779 μs for “Ours” ($3970/779 \approx 5.1$). This claim is accurate. Similarly, the FLOPs reduction claim of “28.4×” in the Abstract is consistent with the complexity analysis in Section 3.3, assuming the parameters $k = 16$ and $B_k = 128$ are used as stated.

Conclusion The paper’s core scientific claims are largely supported by the data, but the phrasing of “matching” performance and the necessity of the “forced local block” mechanism requires refinement to align precisely with the provided ablation studies and long-context results. The citation for hybrid architectures also warrants verification.

Action items.

- **[writing]** The claim that MSA ‘matches GQA on benchmarks’ (Abstract, Intro) is contra-

dicted by Table 2 (HELMET-128K), where MSA-CPT scores 45.93 vs. Full Attention’s 46.53. The text must clarify that performance is ‘comparable’ or ‘statistically indistinguishable’ rather than ‘matching,’ or explicitly note the small degradation in long-context retrieval.

- **[writing]** The claim that the Index Branch ‘always retaining the most recent block’ (Intro) is not fully supported by the ablation in Appendix A.2 (Table: Forced Sink & Local Selection), which states that ‘Removing forced sink/local selection has little effect on quality.’ The text should reconcile this by noting that while the mechanism is present, the model learns to attend to local blocks naturally, making the explicit forcing less critical than claimed.
- **[writing]** The citation \citep{minimax01} in Section 5 is used to support ‘Hybrid stacks...interleave linear and full-attention blocks.’ However, the bib entry for minimax01 describes ‘Lightning Attention’ (a linear attention variant), not necessarily a hybrid stack interleaving linear and full attention. Verify if the citation accurately supports the specific claim of ‘interleaving’ or if a different citation (e.g., a specific hybrid model paper) is needed.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure provides a clear visual overview of the architecture with distinct branches and a helpful legend for the attention masks. However, the caption contains significant text errors, including a missing subject and a broken mathematical formula.

Figure 2

Figure 2 visually fails to support its caption’s claim of distinct long-range patterns across GQA groups, as all four heads show nearly identical attention structures. Axis and colorbar labels are minimally clear but lack contextual precision.

Figure 3

Figure 3 effectively visualizes the mean attention score on the first token for heads in Layer 4 and Layer 24. The bar charts are clearly labeled with values, axes are defined, and the visualization directly supports the caption’s claim of a pervasive attention sink effect.

Figure 4

The figure is visually clear but contains a critical contradiction between the title/legend (GQA Baseline) and the caption (Full-Attention baseline), which misrepresents the reference point for the reported deltas.

Figure 5

The figure effectively illustrates the difference in gradient stability between the two methods, but the truncated x-axis on the gradient norm plot and the premature termination of the ‘No gradient detach’ loss curve limit the ability to assess long-term training behavior.

Figure 6

Figure 6 effectively communicates the benchmark results with clear axes, units, and a legend distinguishing the two conditions. The four subplots are well-organized, and the visual trend supports the caption’s claim that detaching the gradient improves scores.

Figure 7

The figure effectively visualizes the entropy drop and recovery described in the caption, but the y-axis label ‘Value’ is generic and incorrect, and the plot lacks a legend to define the plotted lines.

Figure 8

The figure clearly displays the performance comparison between ‘warmup’ and ‘w/o warmup’ across five benchmarks with readable axes and legends. However, the caption contains a grammatical error and lacks specific context regarding the model or method being evaluated.

Figure 9

The figure effectively visualizes the shift in attention mass between the first token and the learnable sink across different heads. However, the x-axis labels are missing from the top row, and the absence of orange bars in the top plots is visually ambiguous without explicit zero-value indicators.

Figure 10

Figure 10 is a clear and well-constructed line chart comparing perplexity across four benchmarks. The axes are labeled with units, the legend distinguishes the two methods clearly, and the caption accurately reflects the visual data showing no consistent advantage for the learnable sink.

Figure 11

The figure effectively compares the proposed method against a sliding-window baseline across six benchmarks, showing consistent perplexity improvements. However, the caption contains a missing subject in the comparison phrase, and the y-axis label is slightly redundant.

Figure 12

The figure effectively visualizes the training loss curves and includes a helpful inset, but the caption is too brief to stand alone, and the inset's axis labels are too small to read clearly.

Action items.

- **[writing]** Figure 1: The caption reads ‘Overview of .’ with a missing model name or subject immediately following ‘of’, making the sentence incomplete.
- **[writing]** Figure 1: The caption contains a broken mathematical expression ‘a set $\$ofk\$$ key blocks’ where the variable defining the set size is missing between the dollar signs.
- **[science]** Figure 2: The caption claims ‘four GQA groups’ produce ‘different long-range selection patterns,’ but the four subplots (Head 0–3) show nearly identical diagonal and sink patterns with no visible differentiation in long-range behavior.
- **[writing]** Figure 2: The y-axis label ‘Query Block’ and x-axis label ‘Key Block’ are present, but the colorbar label ‘Prob’ is ambiguous without units or context (e.g., normalized probability? attention score?).
- **[science]** Figure 4: The title ‘Capability Delta Relative to GQA Baseline’ contradicts the caption ‘Evaluation-score deltas relative to the Full-Attention baseline’; the baseline definition must be consistent.
- **[science]** Figure 4: The legend includes ‘GQA Baseline’ (dashed line), but the caption states the deltas are relative to a ‘Full-Attention baseline’; the legend entry should match the caption’s baseline definition.
- **[science]** Figure 5: The ‘No gradient detach’ (orange) curve in the LM Loss plot terminates abruptly around step 2500, while the ‘Gradient detach’ (teal) curve continues to step 8500. This missing data prevents a full comparison of training stability and loss convergence over the entire reported range.
- **[science]** Figure 5: The x-axis for the ‘Grad Norm’ plot is truncated at ~768 steps, whereas the ‘LM Loss’ plot extends to 10,000 steps. Since gradient spikes are the key phenomenon being illustrated, the short x-axis range makes it impossible to verify if spikes persist or recur later in training.
- **[writing]** Figure 7: The y-axis is labeled ‘Value’ instead of ‘Entropy’, which contradicts the figure title and caption.
- **[writing]** Figure 7: The legend defining the teal line (Main Branch) and the dashed red line (baseline) is missing from the plot area.

- **[writing]** Figure 8: The caption contains a grammatical error (‘Evaluation results of with and without...’) and fails to specify the model or method being evaluated.
- **[writing]** Figure 8: The x-axis label ‘Tokens (Billion)’ is ambiguous; it is unclear if this represents total training tokens, context length, or model size.
- **[writing]** Figure 9: The x-axis tick labels (512, 2048, etc.) are present only on the bottom row of plots, making the top row (Head 4, Head 5) ambiguous regarding the query positions shown.
- **[writing]** Figure 9: The top row plots (Head 4, Head 5) display only blue bars (‘First token’) but lack the orange bars (‘Learnable sink’) seen in the bottom row, which may confuse readers about whether the sink is absent or simply has zero value.
- **[writing]** Figure 11: The caption contains a grammatical error (‘comparison between and a FLOP-matched...’), missing the name of the proposed method (likely ‘MiniMax Sparse Attention’ or ‘MSA’) which is only identifiable via the legend.
- **[writing]** Figure 11: The y-axis label ‘Perplexity (PPL)’ is redundant; ‘Perplexity’ is sufficient as PPL is the standard abbreviation.
- **[writing]** Figure 12: The caption ‘LM loss’ is insufficient; it fails to describe the comparison between ‘Full Attention’ and ‘MSA-PT’ shown in the legend or the specific training context.
- **[science]** Figure 12: The inset plot’s x-axis labels (2.950, 2.975, 3.000) are illegible and lack units, making it impossible to verify the scale or the specific range of the zoomed-in region.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and hardware jargon that are not defined at their first occurrence, creating barriers for non-specialist readers. In Section 1, the term “MoE” is used to describe the model architecture without defining it as “Mixture of Experts.” Similarly, Section 2.1 introduces “GQA” (Grouped-Query Attention) and “KV heads” without expanding the “KV” acronym, assuming prior knowledge of Transformer internals.

In the kernel design section (Section 5), the text becomes particularly dense with hardware-specific terminology. Section 5.2 mentions handling “hot tiles without atomics” without defining “atomics” as atomic memory operations, and later refers to “128x128 MMAs” without explaining that MMA stands for Matrix Multiply-Accumulate. Section 5.3 introduces “LSE computation” (Log-Sum-Exp) without expansion. While these terms are standard for systems researchers, the paper’s claim of a “minimal, scalable” approach suggests it should be accessible to a broader ML audience. The authors should ensure every acronym is spelled out upon first use and consider

replacing highly specific hardware jargon (like “MMA” or “atomics”) with more descriptive plain English where possible to improve clarity.

Action items.

- **[writing]** The manuscript relies heavily on domain-specific acronyms and hardware jargon that are not defined at their first occurrence, creating barriers for non-specialist readers. In Section 1, the term “MoE” is used to describe the model architecture without defining it as “Mixture of Experts.” Similarly, Section 2.1 introduces “GQA” (Grouped-Query Attention) and “KV heads” without expanding the “KV” acronym, assuming prior knowledge of Transformer internals. In the kernel design section (Section 5), t

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The manuscript presents a coherent architecture for MiniMax Sparse Attention (MSA), but several logical gaps exist between the stated mechanisms and the derived efficiency claims.

First, the central efficiency claim of a 28.4x reduction in per-token compute (Abstract, Sec 1) is not fully supported by the complexity analysis in Sec 4.3. The analysis correctly identifies the Index Branch cost as $O(N^2)$ (specifically $H_{kv}d_{idx}N^2$). At a context length of 1M, this quadratic term is significant. The paper asserts the reduction without explicitly demonstrating that the Index Branch cost is negligible compared to the savings in the Main Branch, or clarifying that the 28.4x figure applies only to the Main Branch operations. Without this distinction, the claim that the *total* attention compute is reduced by this factor is logically unsupported.

Second, the mechanism for block selection contains a definitional ambiguity. Section 3.1 and Algorithm 1 describe the Index Branch scoring as a dot product ($Q_{idx}K_{idx}^T$) followed by TopK selection. However, Section 3.1 also states the Index Branch “aggregates to block level via max-pooling.” It is unclear if the max-pooling is applied to the dot product scores *before* the TopK selection, or if the TopK is applied to the raw dot products and then pooled. If the latter, the “max-pooling” description is misleading; if the former, the algorithm should explicitly show the pooling step before the TopK operation. This ambiguity obscures the causal link between the input tokens and the selected blocks.

Finally, the conclusion in Appendix e001 regarding the removal of the Index Branch value head relies on a “mixed results” argument. The authors observe that removing the head improves some benchmarks (General) while degrading others (Math, Code). Concluding that the head is “not critical” and can be dropped for efficiency ignores the logical implication that the head provides specific, non-redundant capacity for certain reasoning tasks. A more rigorous logical step would be to justify why the efficiency gain outweighs the specific degradation in math/code capabilities, rather than dismissing the trade-off as negligible.

Action items.

- **[writing]** The manuscript presents a coherent architecture for MiniMax Sparse Attention (MSA), but several logical gaps exist between the stated mechanisms and the derived efficiency claims. First, the central efficiency claim of a 28.4x reduction in per-token compute (Abstract, Sec 1) is not fully supported by the complexity analysis in Sec 4.3. The analysis correctly identifies the Index Branch cost as $O(N^2)$ (specifically $H_{kv}d_{idx}N^2$). At a context length of 1M, this quadratic term is signifi

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper exhibits a tendency to overstate the universality of its performance preservation claims, particularly in the Abstract and Conclusion. While the authors claim the method “performs on par with GQA” (Abstract) and “preserves capabilities” (Conclusion), the empirical evidence in Table 2 (HELMET-128K) and Table 1 (GSM8K, MMLU) reveals a nuanced picture where performance is not uniformly preserved. Specifically, the HELMET-128K Overall score drops by 0.60 points, and GSM8K drops by 1.35 points in the CPT setting. While these drops are small, the absolute phrasing “on par” and “preserves” suggests a lack of degradation that the data does not fully support. The authors should qualify these claims to acknowledge the specific benchmarks where slight degradation occurs, distinguishing between “general reasoning” (where performance is stable) and “long-context retrieval” (where trade-offs exist).

Furthermore, the claim of a “28.4x” reduction in per-token attention compute (Abstract, Sec 5.3) is technically accurate regarding FLOPs but risks misleading readers regarding the *effective* efficiency gain if the model requires more tokens or steps to converge due to the slight performance dip. The paper does not explicitly state whether the “on par” claim holds under FLOP-normalized conditions (i.e., if the GQA baseline were scaled down to match MSA’s compute). Without this normalization, the claim that the method achieves massive speedups *while* maintaining performance is slightly overreaching, as it implies a Pareto improvement that might not exist if the baseline were adjusted for compute. The text should clarify that the comparison is against a full-compute GQA baseline, not a compute-matched one.

Finally, the assertion that the Index Branch value head is “not critical” (Appendix e001) is based on a comparison where the “No-value” variant actually outperforms the “With-value” variant on several key benchmarks (MMLU, MMLU-Pro, BBH). While the authors conclude the head is dropped for efficiency, the data suggests it may be detrimental to performance in some contexts. The conclusion that it is merely “not critical” understates the potential negative impact of the value head on certain tasks, which should be framed more carefully as a performance-efficiency trade-off rather than a neutral removal.

Action items.

- **[writing]** The claim of ‘on par with GQA’ (Abstract, Sec 3) is overreaching given Table 2

shows a -0.60 point drop on HELMET-128K Overall. The authors must qualify this claim to reflect that performance is comparable on general benchmarks but slightly degraded on specific long-context retrieval metrics.

- **[writing]** The abstract states the model ‘performs on par with GQA’ while simultaneously claiming a 28.4x FLOP reduction. The paper does not provide a direct FLOP-normalized comparison to prove that the performance is maintained *at the same compute budget*, only at the same parameter count. This conflates efficiency with capability and overstates the efficiency gain’s impact on quality.
- **[writing]** The conclusion claims the method ‘preserves capabilities’ at 109B scale. However, Table 2 shows a 2.40 point gain on HELMET ICL but a 0.60 point loss on Overall, and Table 1 shows mixed results (e.g., -1.35 on GSM8K in CPT). The text should avoid absolute preservation claims and instead report the specific trade-offs observed.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript presents a technical architecture for efficient sparse attention (MSA) and reports performance on standard benchmarks. From a safety and ethics perspective, the paper does not raise immediate red flags regarding dual-use risks, data privacy, or human subject research.

Data Privacy and Consent: The paper describes training on a 109B-parameter model with 3T tokens (Section 4) and continued pretraining on 140B tokens. While the specific composition of the training corpus is not detailed in the text, the authors cite standard public benchmarks (MMLU, GSM8K, RULER, etc.) for evaluation. There is no indication of the use of private, personally identifiable information (PII), or non-consensual data scraping in the methodology described. The model weights are released on Hugging Face, and the code on GitHub, which is standard practice for open research. No IRB or IACUC approval is required as the work involves computational modeling on existing datasets, not human or animal subjects.

Dual-Use and Harm Potential: The proposed method (MSA) is an optimization technique for Large Language Models (LLMs) aimed at reducing computational cost and enabling longer context windows. While more efficient models can theoretically lower the barrier to deploying powerful AI systems, the paper does not introduce new capabilities for generating harmful content, bypassing safety filters, or conducting cyberattacks. The efficiency gains are architectural and apply to the inference/training process generally. The authors do not claim the model is designed for, or capable of, specific high-risk tasks (e.g., biological weapon design, autonomous weapon guidance) that would necessitate a dual-use risk assessment beyond standard LLM safety protocols.

Transparency and Reproducibility: The authors provide links to the model weights

(Hugging Face) and inference kernels (GitHub), facilitating external verification of the claims. The methodology, including the KL alignment loss and block selection strategy, is described with sufficient detail (Section 3, Algorithm 1) to allow for independent reproduction of the training dynamics. The ablation studies (Appendix) further support the validity of the design choices.

Conclusion: The paper adheres to standard ethical norms for AI research. It does not involve human subjects, does not appear to utilize private data without consent, and the technology described does not inherently increase the risk of harm beyond the baseline risks associated with general-purpose LLMs. No action items are required regarding safety or ethics.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling architecture (MSA) with significant efficiency claims, but the scientific evidence supporting the robustness of these claims requires strengthening.

1. Statistical Rigor and Variance: The central claims of massive speedups (14.2x prefill, 7.6x decoding) and FLOP reduction (28.4x) at 1M context (Abstract, Sec 6.3) are presented as single-point measurements. In high-performance computing, especially with custom kernels on H800s, variance due to thermal throttling, memory bandwidth contention, or non-deterministic scheduling can be significant. The manuscript lacks error bars, standard deviations, or confidence intervals. Without multiple runs (e.g., $n=5$) or a clear explanation of how the measurement was stabilized, it is difficult to rule out that the reported speedups are outliers or artifacts of specific hardware states.

2. Baseline Definition and Confounding Variables: In Table 2 (Sec 6.3), the “Full” column is compared against MSA-PT and MSA-CPT. The text states MSA-PT is “trained from scratch” and MSA-CPT is “continued pretraining from GQA checkpoint.” However, it is ambiguous whether the “Full” baseline is a model trained from scratch with full attention or a GQA model with full attention. If the “Full” baseline is a different architecture (e.g., standard Multi-Head Attention vs. GQA), the claim that MSA is “on par” is confounded by the architectural difference between MSA and the baseline. The evidence must explicitly confirm that the “Full” baseline uses the exact same GQA configuration (4 KV heads, 64 Q heads) but with dense attention to isolate the effect of sparsity.

3. Ablation Interpretation: The ablation study in Table 4 (Appendix) regarding the Index Branch value head shows mixed results: the “No-value” variant outperforms “With-value” on MMLU (+0.9) and ARC Challenge (+0.2) but underperforms on GSM8K (-1.2) and MathVista (-1.6). The authors conclude the value head is “not critical” because the differences are “small and benchmark-dependent.” However, without statistical significance testing (e.g., paired t-tests across seeds), these differences could be noise. Furthermore, the claim that the value head is “not critical” contradicts the earlier statement that it helps the model “begin sparse training

from step zero.” The evidence suggests the value head may be critical for *stability* or *specific task types* (math) even if average performance is similar. The conclusion should be nuanced to reflect this task-dependent variance.

4. Contradictory Stability Claims: Section 3.1 states the local block is “always included” to “ensure stability.” However, the ablation in Table 5 (Appendix) shows that “Removing forced sink/local selection has little effect on quality.” If the forced selection is truly necessary for stability, removing it should degrade performance, particularly in long-context scenarios. The fact that it has “little effect” suggests the model learns to attend to local blocks naturally, rendering the forced mechanism redundant. This contradicts the design rationale and weakens the evidence for the necessity of the forced local block mechanism. The authors should clarify if the “little effect” applies to long-context retrieval (RULER) or only short-context benchmarks.

Action items.

- **[science]** The claim of 28.4x FLOP reduction and 14.2x/7.6x speedups at 1M context (Abstract, Sec 6.3) lacks statistical validation. Provide standard deviations or confidence intervals across multiple runs or seeds to rule out variance from hardware noise or non-deterministic kernels.
- **[science]** The ‘Full Attention’ baseline in Table 2 (Sec 6.3) is not explicitly defined as a model trained with full attention from scratch or a GQA model with full attention. If the baseline is a GQA model with full attention, the comparison is valid; if it is a different architecture, the claim of ‘on par’ performance is confounded by architectural differences.
- **[science]** The ablation study in Table 4 (Appendix) shows mixed results for the ‘No-value’ variant (e.g., +0.9 on MMLU, -1.2 on GSM8K). The conclusion that the value head is ‘not critical’ is weak without a statistical test (e.g., paired t-test) to determine if these differences are significant or within noise.
- **[science]** The paper claims the Index Branch ‘always includes the local block’ (Sec 3.1) but the ablation in Table 5 (Appendix) shows ‘Removing forced sink/local selection has little effect’. This contradicts the stability claim. Clarify if the ‘little effect’ refers to perplexity only or also to long-context retrieval, and provide evidence for the stability claim.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The manuscript presents a novel sparse attention mechanism (MSA) with compelling efficiency claims. However, the statistical rigor of the empirical evaluation is insufficient to fully support the claims of “on par” performance or specific marginal improvements.

Lack of Uncertainty Quantification: Throughout Section 5 (Experiment), specifically in Tables 1 and 2, the authors report single-point benchmark scores (e.g., MMLU 67.2 vs 67.0,

RULER-128K +0.12). In large-scale LLM training, performance variance across random seeds or evaluation batches can be significant. The absence of standard errors, confidence intervals, or p-values makes it impossible to determine if the observed differences are statistically significant or merely noise. For instance, the claim that MSA “matches” Full Attention on MMLU (67.2 vs 67.0) is plausible, but without error bars, the claim that it “excels” in math (26.3 vs 25.9) is statistically unsupported.

Reproducibility of Efficiency Metrics: Section 5.3 and Table 1 report specific speedup factors (e.g., 14.2x prefill). While the kernel latency table (Table 1) shows consistent trends, the end-to-end speedup is presented as a single measurement. To ensure reproducibility and robustness, the authors should report the number of independent runs and the standard deviation for these speedup metrics.

Recommendation: The authors should either re-run experiments with multiple seeds to provide mean $\pm$ standard deviation for all benchmark scores or explicitly state that results are from a single run and frame the conclusions as “point estimates” rather than definitive statistical improvements. Without this, the claims of performance parity and specific gains remain anecdotal.

Action items.

- **[science]** Section 5.2 and Table 1 report benchmark scores (e.g., MMLU 67.2 vs 67.0) without standard errors, confidence intervals, or p-values. Given the single-run nature of large-scale training, authors must clarify if these are single seeds or averages, and provide uncertainty estimates to support claims of statistical equivalence.
- **[science]** Table 1 and Table 2 present differences (e.g., +0.12 on RULER-128K) as definitive improvements. Without multiple independent runs or a statistical significance test, these marginal gains may be within the noise floor of the evaluation pipeline. Authors should either run multiple seeds or explicitly frame these as point estimates without claiming superiority.
- **[science]** The efficiency claims in Section 5.3 (14.2x speedup) rely on single measurements on H800. While kernel latency (Table 1) shows consistent gains, the end-to-end speedup lacks variance reporting. Authors should report the number of runs and standard deviation for the speedup metrics to ensure reproducibility.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a high level of technical clarity and generally adheres to standard academic writing conventions. The logical flow from the problem statement (quadratic cost) to the proposed solution (MSA) and its evaluation is coherent. The abstract effectively summarizes the key contributions and results.

However, there are a few minor areas where phrasing could be tightened to improve precision and readability. In Section 5, the term “Programmatic Dependent Launch” is used; while likely referring to a specific CUDA mechanism, the phrasing is slightly non-standard and could be clarified to ensure immediate understanding by a broader systems audience. Additionally, in the Appendix (e001), the use of the word “recipe” to describe the training methodology is slightly informal; “methodology” or “procedure” would be more appropriate for a formal publication. Finally, in Section 3, the term “visible key tokens” in the Index Branch description could be more explicitly defined as “causally visible” to avoid any ambiguity regarding the scope of the attention mechanism.

Overall, the writing is strong, and these are minor stylistic adjustments rather than fundamental issues with comprehension. The paper is well-structured and the arguments are presented clearly.

Action items.

- **[writing]** In Section 5 (Kernel Design), the phrase ‘Programmatic Dependent Launch’ appears to be a non-standard or awkward phrasing for ‘Programmatic Dependent Launch’ (if referring to CUDA streams) or ‘Programmatic Dependency’. Verify the standard terminology for the specific CUDA feature used to avoid confusion.
- **[writing]** In the Appendix (e001), the sentence ‘This suggests that the main role of O_idx in the earlier recipe was to provide an additional early training signal...’ uses ‘recipe’ in a context that feels slightly informal for a technical paper. Consider replacing with ‘methodology’, ‘architecture’, or ‘training procedure’.
- **[writing]** In Section 3 (Method), the description of the Index Branch states it ‘Scores visible key tokens’. The term ‘visible’ is ambiguous here; does it mean ‘causally visible’ (i.e., past and current tokens) or ‘visible to the specific query’? Clarify to ensure precise technical meaning.